Lecture 36: The Regulatory Path to Approval
The Oklo Aurora Case Study

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Introduction: The Traditional Path to Approval

For over 50 years, getting a new commercial nuclear reactor approved in the United States meant navigating the **Nuclear Regulatory Commission (NRC)**. This framework, codified in 10 CFR 50 and 52, relies heavily on prescriptive, deterministic safety analysis and one non-negotiable feature: a robust mechanism for containment of radioactivity, typically a reinforced concrete **Containment Dome**.

Under the NRC "gold standard," you assume a maximum credible accident (like a massive pipe break) and rely on a reinforced concrete structure to trap the radiation. This is highly effective for safety, but the approval process is notoriously cumbersome, expensive, and rigid. For emerging Small Modular Reactor (SMR) startups, the traditional NRC pathway presents a massive financial and bureaucratic wall.

In this lecture, we will examine how one company—Oklo—attempted to navigate this wall, failed, and ultimately pivoted to a completely new regulatory pathway, fundamentally altering their technology in the process.

1 Act I: The Original Aurora and the NRC Rejection

When Oklo first submitted their Combined License Application to the NRC in 2020, the "Aurora" was pitched as a revolutionary, highly innovative piece of technology.

1.1 The Innovative Design (Aurora v1)

The original Aurora design completely abandoned traditional light-water mechanics.

- **Micro-Scale Output:** The original Aurora was a true "microreactor," designed to produce just 4 MW of thermal power (MWt), yielding about 1.5 MW of electricity (MWe). To put this in perspective, the NRC application noted that the entire active core of the Aurora contained roughly the same volume of fuel as a *single fuel assembly* at a large commercial plant like Diablo Canyon.
- **The Fuel:** U-10Zr (metallic uranium-zirconium alloy) using High-Assay Low-Enriched Uranium (HALEU).
- **Heat Removal (Potassium Heat Pipes):** Instead of active coolant pumps or a massive pool of liquid metal, Oklo proposed a "forest" of sealed heat pipes. Each pipe contained a small amount of potassium working fluid operating passively at sub-atmospheric pressure.

As thermal superconductors, the potassium would vaporize at the bottom, travel upward, condense, and return via capillary action, offering redundant and reliable cooling for the tiny core.

- **Power Conversion (sCO_2):** To completely avoid the historic and highly explosive hazards of alkali metal-water reactions (such as potassium or sodium mixing with water), the heat pipes were coupled to a supercritical CO_2 (sCO_2) Brayton cycle rather than a steam turbine.
- **The Catch:** Oklo argued this system was so small, and inherently safe ("walk-away safe"), that they **did not need a containment dome**, proposing that their fuel provided **functional containment** and that they could house their reactor in a standard industrial weather structure (an A-frame building).

1.2 The Economic Motivation (The Square-Cube Law)

Why fight the NRC over the dome? Because of the Square-Cube law. For a large 1,100 MW reactor, a $500 million concrete dome adds roughly $5/MWh to the cost of electricity. But if you scale that down to a tiny 1.5 MWe to 50 MWe SMR, the power output (volume) shrinks much faster than the required surface area of the dome.

$$C \propto P^{2/3}, \qquad \frac{C}{P} \propto P^{-1/3} \tag{1}$$

Scaling from 1,100 MW down to the SMR scale yields a massive unit-cost increase. A containment dome on a micro-reactor could add over $15/MWh, rendering the plant completely economically uncompetitive against natural gas. Eliminating the dome was an economic mandate for Oklo.

1.3 The 2022 NRC Rejection

Oklo attempted to push this dome-less, micro-scale heat-pipe design through the rigid NRC. In 2022, the NRC denied the application.

- **The Reason:** The NRC demanded validated data and bounding analyses on what happens to metallic fuel during degraded heat-removal conditions, and what the quantitative source-term release would be if the cladding failed without a containment dome to catch it. Because Oklo relied on theoretical "functional containment" rather than empirical data, and because their application included gaps in safety analysis, design detail, and accident modeling, the NRC refused to proceed.

2 Act II: The Regulatory Pivot and the 2026 DOE Rewrite

Following the NRC rejection, the regulatory landscape in the US shifted dramatically. Seeking to bypass the NRC bottleneck, startups looked toward the Department of Energy (DOE), which has the authority to authorize experimental reactors on federal sites (like Idaho National Laboratory) outside of standard NRC commercial licensing.

2.1 The July 2026 Deadline and the NPR Analysis

In mid-2025, a presidential executive order mandated that the DOE approve at least three new experimental reactors to achieve criticality by July 4, 2026—an impossibly tight timeline in the nuclear world.

To make this happen, the DOE fundamentally rewrote its safety rules. As analyzed by NPR and ProPublica in early 2026, this rewrite represents a massive shift in regulatory philosophy:

1. **From "Shall" to "May":** The DOE systematically scrubbed prescriptive, mandatory language from its safety orders. Phrases dictating that operators "shall" ensure certain safety margins or that systems "must" survive specific transients were replaced with permissive wording ("may" or "should").

2. **Categorical Exclusions (CE):** The DOE granted advanced nuclear projects broad exemptions from standard National Environmental Policy Act (NEPA) environmental reviews.

3. **Organizational and Workforce Shifts:** Reporting by NPR and other outlets has described a broader shift toward a more rapid, innovation-driven regulatory approach, sometimes characterized as a "move fast and break things" ethos. At the same time, public reporting indicates that the NRC experienced on the order of 200–250 retirements and resignations among inspection and technical staff over a relatively short period, while hiring only ∼60 replacements.

4. **Oversight Capability:** From an engineering oversight perspective, this represents a substantial reduction in experienced personnel responsible for inspection, verification, and enforcement, and may increase reliance on licensee-provided analyses relative to independent federal review. This is particularly problematic in view of all the novel proposed reactor designs.

5. **Change in NRC Licensing:** While DOE can license "experimental reactors", the NRC is still responsible for commercial reactor licensing. In an NRC/DOE memorandum of understanding (August 2025) the NRC is now required to provide expedited commercial licensing to a reactor based on empirical results from DOE licensed prototypes. This circumvents the usual "what if" aspect of NRC evaluation.

3 Act III: The Design Pivot (Aurora v2 & the EBR-II Legacy)

Faced with aggressive DOE timelines for pilot reactors, Oklo realized that getting their novel, unproven potassium heat-pipe and sCO_2 technology built and authorized was mathematically and physically impossible. They needed historically proven technology.

3.1 Pivoting to an EBR-II Derived Design

The "new" Aurora is no longer a true first-of-a-kind (FOAK) technology in its core physics. It is fundamentally based on the Experimental Breeder Reactor II (EBR-II) that successfully operated in Idaho for 30 years.

While the underlying physics are legacy, the engineering scale-up is significant:

- **A Massive Scale-Up:** The new design abandons the tiny 4 MWt microreactor concept. The revised Aurora targets 75 MWe. This represents a massive increase in power output and physical size, fundamentally changing the thermal-hydraulic, decay heat, and source-term calculations from the original EBR-II's 20 MWe output.

- **Breeding vs. Burning:** While EBR-II was designed to breed fuel, Aurora is designed as a "burner" to efficiently consume isotopes.
- **The Fuel Source:** Aurora will run on the literal ashes of its predecessor, utilizing HALEU recovered directly from spent EBR-II fuel.
- **Pool-Type Sodium:** They abandoned the potassium heat pipes for a standard liquid sodium pool and intermediate sodium loop architecture.
- **Steam Turbine:** Because commercial sCO_2 turbines do not readily exist, Oklo signed a contract with Siemens Energy to use a standard, off-the-shelf SST-600 steam turbine to keep capital expenditures manageable.

3.2 The Branding Question: Why Keep the Name?

If the reactor is vastly larger, uses a massive pool of liquid sodium instead of potassium heat pipes, and drives a steam turbine instead of a supercritical CO_2 cycle—why keep the name "Aurora"?

A likely explanation lies in the practical realities of project development and finance. Maintaining continuity in project identity helps avoid renegotiation of power purchase agreements, re-evaluation of early-stage site permits, and potential uncertainty among investors. The persistence of the "Aurora" name reflects how non-technical constraints shape the presentation of engineering systems.

4 Act IV: The Remaining Technical Risk

Oklo successfully pivoted their regulatory approach (from the strict NRC to the permissive DOE) and their design approach. However, they **still refuse to build a containment dome**, relying on the relaxed DOE rules to allow a standard industrial building. This introduces a major, historically documented risk:

4.1 The Secondary Sodium-Water Interface Risk

By reverting to an EBR-II style pool and a standard steam turbine, Oklo brought back the oldest engineering hazard in fast reactors: the sodium-water reaction. If a tube leaks in the steam generator, high-pressure steam will mix with secondary sodium, producing explosive hydrogen gas, highly caustic sodium hydroxide, and massive pressure spikes:

$$2\mathrm{Na} + 2\mathrm{H_2O} \rightarrow 2\mathrm{NaOH} + \mathrm{H_2} + \mathrm{Heat} \tag{2}$$

While the original EBR-II utilized a steel containment dome to seal in the reactor (though not a reinforced concrete pressure dome), Oklo's Aurora proposes a standard industrial weather structure. Licensed under the new DOE regulations, the success of the Aurora pilot will test whether an un-domed sodium plant can truly be operated safely, or if the classic NRC "gold standard" was right all along.

5 Act V: The 2026 Reality Check: Shifting Timelines and the War Chest

Despite the unprecedented relaxation of regulatory hurdles under the DOE, the physical reality of heavy nuclear construction remains a formidable barrier. Oklo's path forward is defined by a dual-track strategy, shifting deadlines, and a massive influx of capital.

5.1 The "Portfolio Strategy" and the Atomic Alchemy Acquisition

A common misconception is that Oklo planned for their flagship 75-MWe Aurora powerhouse to meet the politically mandated July 4, 2026, DOE deadline. In reality, that deadline applies to the DOE program as a whole, mandating that *at least three* selected reactors hit criticality. Oklo knew the complex Aurora wouldn't be ready until 2028.

To secure a viable candidate for the July 4th race, Oklo executed a strategic pivot:

- **Atomic Alchemy Acquisition:** On February 28, 2025, Oklo acquired Y Combinator-backed startup Atomic Alchemy in a $25 million all-stock deal.
- **The Groves VIPR:** This acquisition gave Oklo control of the Groves Isotopes Test Reactor in Texas—a 15-MWt water-cooled Versatile Isotope Production Reactor. Designed to produce medical isotopes rather than power, this simpler reactor is Oklo's actual horse in the July 2026 criticality race, buying them time to focus on their main objective: the Aurora.

5.2 The Financial Reality: High Burn, Deep Pockets

The narrative of Oklo facing an imminent financial cliff ignores their masterful capitalization on market hype. However, the true cost of building first-of-a-kind (FOAK) infrastructure has still shocked the market.

- **The Timeline Shift:** Originally pitched to investors for a 2026/2027 deployment, Oklo officially updated its guidance in late 2025, targeting late 2027 to early 2028 for the Aurora's commercial operation at the Idaho National Laboratory (INL).
- **Missed Earnings and the CapEx Furnace:** Oklo's March 2026 earnings report was a harsh reality check. They reported a substantial EPS miss driven by massive 2026 capital expenditure (CapEx) projections of $350 million to $450 million, plus an operational burn rate of $80 million to $100 million as they juggle multiple reactor sites.
- **The $2.5 Billion War Chest:** Despite the necessarily high burn rate, Oklo is in a uniquely strong economic position. Trading at $10 or lower up to September 2024, Oklo's stock spiked in 2025, reaching an intra-day high of $193 that October. By executing At-The-Market (ATM) stock sales during the late 2025 hype and completing a massive $1.18 billion secondary offering in January 2026, Oklo built a cash war chest of roughly $2.5 billion. They diluted their shareholders significantly to do it (it now trades around $50), but they successfully secured the capital required to survive the "valley of death" of FOAK nuclear construction.

6 Act VI: The Future — The Ohio Campus and the Prototype Loophole

With their short-term survival secured by the Wall Street capital raise, Oklo's long-term valuation now hinges entirely on a single, massive commercial deployment and a highly calculated regulatory gamble.

6.1 The Meta Deal and the Pike County Campus

Oklo's endgame is not building single test reactors in Idaho; it is powering the artificial intelligence boom. In early 2026, Oklo secured a commercial anchor agreement with Meta to provide up to 1.2 gigawatts of power for their data centers.

- **The Customer-Funded Model:** Meta provided an upfront cash prepayment to help Oklo secure nuclear fuel and begin site preparations on a 206-acre private site in Pike County, Ohio.
- **The Reactor Fleet:** To hit the 1.2 GW requirement, Oklo plans to deploy a phased campus of up to 16 interconnected Aurora reactors on the Ohio site, with the first units targeted to come online around 2030.

6.2 The Regulatory Strategy: Bypassing the NRC

Because the Ohio site is private land, Oklo cannot simply build reactors under the Department of Energy's (DOE) relaxed oversight; they must face the notoriously slow Nuclear Regulatory Commission (NRC). To solve this, Oklo is utilizing the newly established legal loophole:

1. **The INL Prototype:** Oklo is currently building the first Aurora at INL under DOE authorization.
2. **Empirical Data:** By operating the INL reactor, they will generate the hard, empirical safety data required by law to prove their novel liquid-metal fast reactor (LMFR) works.
3. **The Expedited Pathway:** Under a newly signed Memorandum of Understanding (MOU) between the DOE and NRC, the NRC is required to establish an expedited commercial licensing pathway for designs that have been successfully tested and authorized by the DOE rather than revisiting risks that have already been addressed in the DOE review. Oklo intends to use the INL data to force an expedited NRC approval for the Ohio campus.

6.3 System-Level Dependency Risk: The LMFR Uptime Paradox

Oklo's entire master plan—the 2030 Ohio timeline, the Meta partnership, and the expedited NRC license—rests on a massive, unproven assumption: that the Aurora prototype at INL will actually work as intended.

Historically, liquid sodium-cooled fast reactors have been plagued by catastrophic reliability issues, leaks, and fires, resulting in abysmal uptime (such as France's Superphénix operating at a mere 14% lifetime capacity factor). If Oklo's first Aurora at INL suffers from these exact same thermal-hydraulic failures, they will not generate the positive empirical data needed for the NRC. Without that data, the expedited MOU pathway closes.

An even more serious worry is that the INL reactor works fine at first, the Ohio facility is built out, and then a design vulnerability is later exposed with disastrous consequences. This presents a latent, systemic risk: an operational failure discovered years into the Ohio build-out could instantly compromise the regulatory basis of the entire deployed fleet.

Engineering Parallel: The Concorde Syndrome

The history of the commercial supersonic transport Concorde offers an instructive engineering parallel. The aircraft operated for decades with an exceptional safety record and no fatal accidents prior to 2000. However, a single event—runway debris causing a tire burst that led to fuel tank rupture and subsequent fire—revealed a previously unrecognized vulnerability shared across the fleet. Although the technical issue was ultimately correctable, the perceived safety of the aircraft collapsed rapidly, and the fleet was retired within a few years due to a combination of safety concerns, economic factors, and declining demand.

Key Engineering Concept: Common-Mode Failure
A common-mode failure is a flaw shared across an entire fleet due to common design or operating assumptions. When triggered, it can simultaneously invalidate the safety case for all units.

In nuclear engineering, a deployed fleet of advanced reactors (e.g., sodium fast reactors) could face a similar challenge. If a previously unrecognized common-mode failure were identified after large-scale deployment, the regulatory and public response could mandate a fleet-wide shutdown even if the problem is technically correctable, and in some cases the shutdown may persist due to regulatory caution and public response.

A real-world example occurred after the Fukushima accident, when Japan shut down its entire fleet of 54 reactors as a precautionary response to a perceived systemic risk. More than a decade later, only a small fraction of these reactors have returned to operation.

7 Details of the Regulatory Shift: From Prescriptive to Discretionary

Following the 2025 executive actions aimed at accelerating experimental reactor deployments, the Department of Energy (DOE) revised portions of its safety and operational rules.

A notable feature of these revisions is a shift in language from prescriptive requirements ("shall," "must") toward more discretionary or performance-based phrasing ("may," "as appropriate"). Such changes do not eliminate safety requirements, but they can alter how those requirements are implemented, verified, and enforced. In this section, we examine several specific rule changes as cited by NPR and analyze their potential engineering implications.

7.1 Case Study 1: The Groundwater Rules

Text of the Rule Change

CURRENT RULES (DOE O 458.1):
PROTECTION OF DRINKING WATER AND GROUND WATER
Ground water must be protected from radiological contamination to ensure compliance with dose limits in the Order and consistent with ALARA process requirements. To this end, DOE must ensure that: (a) Baseline conditions of the ground water quantity and quality are documented; (b) Possible sources of, and potential for, radiological contamination are identified and assessed; (c) Strategies to control radiological contamination are documented and implemented; (d) Monitoring

methodologies are documented and implemented; and (e) Ground water monitoring activities are integrated with other environmental monitoring activities.

NEW RULES (NE O 458.1A):
GROUND WATER
Consideration must be given to avoiding or minimizing potential radiological contamination of groundwater from NE-authorized nuclear facilities and radiological activities, to ensure compliance with public dose limits in this Order. To this end: (i) baseline conditions of the ground water quantity and quality may be considered; (ii) potential sources of radiological contamination may be identified; (iii) strategies to control radiological groundwater contamination may be developed, as appropriate; (iv) monitoring may be implemented, as appropriate, and integrated with other monitoring activities.

Engineering Implication

The revised language removes explicit prescriptive requirements for baseline characterization, monitoring, and documentation, replacing them with discretionary phrasing such as "may be considered" and "as appropriate." It also eliminated any reference to the ALARA principle "As Low as Reasonably Achievable," a rather startling change.

From an engineering and environmental monitoring perspective, baseline data play a critical role in attribution. Without well-characterized pre-operational conditions, it becomes significantly more difficult to distinguish facility-induced contamination from pre-existing or background conditions using standard methods such as trend analysis or plume reconstruction.

Similarly, discretionary monitoring requirements may lead to increased variability in how different facilities implement environmental surveillance programs. While monitoring is not necessarily eliminated, the shift in language transfers greater responsibility to site-specific judgment and regulatory interpretation.

7.2 Case Study 2: Accident Investigations and Dose Thresholds

Text of the Rule Change

CURRENT RULES (DOE O 225.1B):
LOSS OF CONTROL OF RADIOACTIVE MATERIAL
(1) Any single accident that results in: (a) A general employee exceeding any of the external dose limits in 10 C.F.R. Part 835.202, Occupational Dose Limits for General Employees, by a factor of two or more.

NEW RULES (NE O 225.1):
LOSS OF CONTROL OF RADIOACTIVE MATERIAL
(1) Any single accident that results in: a. A general employee exceeding any of the external dose limits in 10 C.F.R. Part 835.202, Occupational Dose Limits for General Employees, by a factor of four or more.

Health Physics Implication

The standard occupational dose limit for radiation workers is 5 rem/year. Under the previous rule, a single-event exposure exceeding 10 rem triggered formal investigation; the revised rule raises this threshold to 20 rem.

From a biological standpoint, an acute dose of 20 rem (0.2 Sv) is generally below the threshold for acute radiation syndrome (ARS), which typically begins near 100 rem. However, such a dose is well above normal operational expectations and represents a significant radiological control event.

From an engineering perspective, exposures of this magnitude typically indicate a breakdown in one or more layers of radiological protection, such as shielding, procedural controls, or access restrictions. The lower investigation threshold historically served as an early-warning trigger to identify and correct such failures before they could escalate.

Raising the threshold does not change dose limits themselves, but it does alter when formal investigation is required. This may delay systematic root-cause analysis in certain scenarios, depending on how the revised criteria are implemented.

7.3 Case Study 3: Physical Security and Protective Forces

Text of the Rule Change

CURRENT RULES (DOE O 473.2A):
PERSONAL PROTECTIVE ARMOR
Personal protective armor must be issued for all SPOs [Security Police Officers]. (1) Protective armor for SPOs must provide Type III protection, at a minimum... (2) Protective armor must be worn at all times by: (a) SPOs assigned posts that interact with the general public... (b) SPOs at sites with PL 1-4 assets. The ODFSA may waive this requirement for specific posts... This waiver must be documented.

NEW RULES (NE O 470.1):
(A) PERSONAL PROTECTIVE ARMOR.
The contractor will: i. Issue personal protective armor for all SPOs as identified or required through local analysis. ii. Ensure issued protective armor for SPOs provides Type III protection, at a minimum... iii. Wear protective armor as directed by the ODSA.

Operational Implication

The revised rule shifts decision-making authority regarding protective posture from a centralized requirement to site-specific analysis. Rather than mandating universal use of protective armor in defined situations, the new framework allows deployment decisions to be based on local threat assessments.

From a security engineering perspective, nuclear facilities are traditionally designed using a *defense-in-depth* approach, incorporating multiple layers of protection including physical barriers, armed response, and deterrence measures. Standardization across sites can contribute to predictable baseline security levels. Simply put, it helps make a nuclear facility a "hard target," contributing to deterrence by increasing the perceived difficulty and risk of adversary actions.

Introducing site-specific discretion may allow operators to tailor security posture to local conditions, but it may also lead to variability in implementation across facilities. The engineering question becomes one of consistency: whether decentralized decision-making maintains equivalent levels of protection across a distributed fleet of reactors.

7.4 Case Study 4: Protection of Biota and Ecological Impact

Text of the Rule Change

CURRENT RULES (DOE O 458.1):
PROTECTION OF BIOTA.
(1) Radiological activities that have the potential to impact the environment must be conducted in a manner that protects populations of aquatic animals, terrestrial plants, and terrestrial animals in local ecosystems from adverse effects due to radiation and radioactive material released from DOE operations.
(2) When actions taken to protect humans from radiation and radioactive materials are not adequate to protect biota then evaluations must be done to demonstrate compliance with paragraph 4.j.(1) of this Order in one or more of the following ways: (a) Use DOE-STD-1153-2002, A Graded Approach for Evaluating Radiation Doses to Aquatic and Terrestrial Biota. (b) Use an alternative approach to demonstrate that the dose rates to representative biota populations do not exceed the dose rate criteria in DOE-STD-1153-2002, Table 2.2. (c) Use an ecological risk assessment to demonstrate that radiation and radioactive material released from DOE operations will not adversely affect populations within the ecosystem.

NEW RULES (NE O 458.1):
BIOTA
Consideration may be given to avoiding or minimizing, if practical, potential adverse impacts to aquatic animals, terrestrial plants, and terrestrial animals in local ecosystems from radiation and releases of radioactive material, using a graded approach.

Environmental Engineering Implication

The revised language removes the explicit requirement for a standardized, quantitative evaluation framework for protection of non-human biota. Under the previous rule, if baseline protections for human health were not sufficient to protect the surrounding ecosystem, facility designers were required to demonstrate compliance using defined methods, such as the dose rate criteria in DOE-STD-1153-2002 or a formal ecological risk assessment.

In contrast, the new rule replaces these prescriptive requirements with discretionary language such as "consideration may be given" and "if practical," without reference to specific dose limits or required analytical methods. This represents a shift from a compliance-based framework toward a more performance-based or judgment-based approach.

From an engineering perspective, the introduction of "if practical" implicitly incorporates feasibility, cost, and project constraints into decisions about environmental protection. As a result, the extent to which ecological impacts are quantitatively evaluated or mitigated may vary across facilities, depending on site-specific decisions and regulatory interpretation.

The removal of explicit dose criteria as a required design check may also reduce the likelihood that protection of non-human biota serves as a primary driver for the inclusion of dedicated monitoring or mitigation systems. While environmental protection is not eliminated, it may become less uniformly defined and less explicitly quantified than under the previous framework.

More broadly, this change illustrates a recurring theme in the revised regulatory structure: a transition from prescriptive, standardized requirements toward flexible, case-dependent implementation. The engineering implications of this shift depend not only on the written rules, but on how rigorously they are applied in practice.

7.5 Summary

Across these examples, the common feature is not the removal of safety requirements *per se*, but a shift in how those requirements are specified. Prescriptive rules provide uniformity and enforceability, while discretionary rules allow flexibility and adaptation.

From an engineering standpoint, this shift transfers greater responsibility to:

- Site operators, who must interpret and implement requirements
- Regulators, who must evaluate compliance under less rigid criteria
- Designers, who must account for a wider range of acceptable implementations

The long-term impact of this transition will depend not only on the written rules, but on how they are applied in practice. In particular, greater flexibility may lead to increased variability in how safety margins and environmental protections are implemented across different projects.

“If you lived downwind of the Aurora Powerhouse, would you prefer a visually appealing glass-walled reactor building with no large containment structure, or a reinforced containment dome designed for worst-case accidents—and why?”

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